

Chao Xu

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Research Interests

Combinatorial Optimization · Computational Geometry · Algorithms

Education

- 2013-2018 PHD in Computer Science, University of Illinois at Urbana-Champaign
Advisors: Karthik Chandrasekaran and Chandra Chekuri.
- 2009-2013 BS in Mathematics and Applied Mathematics & Statistics with minor in Computer Science, Stony Brook University

Employment

- Jan.2023-
now **Associate Professor, UESTC**, Chengdu, China.
School of Computer Science and Engineering.
- Jan.2022-
Dec.2022 **Assistant Professor, UESTC**, Chengdu, China.
School of Computer Science and Engineering.
- Mar.2020-
Mar.2021 **Software Engineer, Voleon**, Berkeley, CA, USA.
Research Engineering.
- Sep.2019-
Mar.2020 **Senior Software Engineer, Grab**, Bellevue, WA, USA.
Grab Artificial Intelligence Accelerator.
- Jun.2018-
Aug.2019 **Research Scientist, Yahoo! Research**, New York, NY, USA.
Scalable Machine Learning Group.
- Feb.-Aug.
2013 **Software Engineer, Google**, Mountain View, CA, USA.
Google Analytics Backend.

Visiting Positions

- Jan.-June.
2023 **Visiting Researcher, Tepper School of Business, Carnegie Mellon University**, Pittsburgh, USA.
Hosted by Benjamin Moseley.
- Jun.-Aug.
2017 **Visiting Researcher, National Institute of Informatics**, Tokyo, Japan.
Hosted by Ken-ichi Kawarabayashi.
- Jun.-Aug.
2015 **Visiting Scholar, New York University**, New York, USA.

Conference Publications¹

- 2026* .
- 2025 Y. Chen, K. Shi, and C. Xu. An Optimal Algorithm for the Stack Crane Problem on Fixed Topologies. In **ISAAC 2025** (*International Symposium on Algorithms and Computation*), 2025.
- 2025* Y. Hao, Y. Zhou, C. Xu, and Z.-H. Fu. Efficient Branch-and-Bound for Submodular Function Maximization under Knapsack Constraint. In **ECAI 2025** (*European Conference on Artificial Intelligence*), 2025.
- 2025* Y. Cong, C. Xu, and Y. Zhou. Large-scale Trade-off Curve Computation for Incentive Allocation with Cardinality and Matroid Constraints. In **IJCAI 2025** (*Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*), 2025.
- 2024 S. Liu and C. Xu. On the Congruency-Constrained Matroid Base. In **IPCO 2024** (*25th Conference on Integer Programming and Combinatorial Optimization*), 2024.
- 2024* R. Bai, C. Xu, C. Xu, and R. Zhang. Resource-Limited Network Security Games with General Contagious Attacks. In **COCOON 2024** (*International Computing and Combinatorics Conference*), 2024.
- 2024* K. Shi and C. Xu. Almost Optimum ℓ -Covering of Z_n . In **COCOON 2024** (*International Computing and Combinatorics Conference*), 2024.
- 2024* D. Lv, Y. Song, and C. Xu. Nested and Interlaced Ticketing for Multiple Travelers. In **IJTCS-FAW 2024** (*International Joint Conference on Theoretical Computer Science – Frontier of Algorithmic Wisdom*), 2024.
- 2023 J. Wang, C. Xu, and S. Zhou. Solving systems of linear equations through zero forcing set. In W. Wu and G. Tong, editors, *Computing and Combinatorics*, pages 353–365, Cham, 2024. Springer Nature Switzerland.
- 2023 T. Hirayama, Y. Liu, K. Makino, K. Shi, and C. Xu. A Polynomial Time Algorithm for Finding a Minimum 4-Partition of a Submodular Function. In **SODA 2023**.
- 2022* P. Yu, C. Xu, A. Bifet, and J. Read. Linear TreeShap. In **NeurIPS 2022, Oral**.
- 2022 C. Beideman, K. Chandrasekaran, C. Chekuri, and C. Xu. Approximate Representation of Symmetric Submodular Functions via Hypergraph Cut Functions. In **FSTTCS 2022**.
- 2022* J. Zhao, M. Xiao, and C. Xu. Improved Approximation Algorithms for the Traveling Tournament Problem. In S. Szeider, R. Ganian, and A. Silva, editors, *47th International Symposium on Mathematical Foundations of Computer Science (MFCS 2022)*, volume 241 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 83:1–83:15, Dagstuhl, Germany, 2022. Schloss Dagstuhl – Leibniz-Zentrum für Informatik.
- 2020 C. Beideman, K. Chandrasekaran, and C. Xu. Multicriteria Cuts and Size-Constrained k -Cuts in Hypergraphs. In J. Byrka and R. Meka, editors, *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques (APPROX/RANDOM 2020)*, volume 176 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 17:1–17:21, Dagstuhl, Germany, 2020. Schloss Dagstuhl–Leibniz-Zentrum für Informatik.

¹By convention in theoretical computer science and mathematics, author orders of all papers are alphabetical.

* the author order is determined by contribution.

- 2019 C. Chekuri, K. Quanrud, and C. Xu. LP Relaxation and Tree Packing for Minimum k -cuts. In J. T. Fineman and M. Mitzenmacher, editors, *2nd Symposium on Simplicity in Algorithms (SOSA 2019)*, volume 69 of *OpenAccess Series in Informatics (OASICS)*, pages 7:1–7:18, Dagstuhl, Germany, 2018. Schloss Dagstuhl–Leibniz-Zentrum fuer Informatik.
- 2018 K. Chandrasekara, C. Xu, and X. Yu. Hypergraph k -cut in randomized polynomial time. In *Proceedings of the Twenty-Ninth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 1426–1438, 2018.
- 2017 K. Bérczi, K. Chandrasekaran, T. Király, E. Lee, and C. Xu. Global and Fixed-Terminal Cuts in Digraphs. In *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques (APPROX/RANDOM 2017)*, volume 81 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 2:1–2:20, Dagstuhl, Germany, 2017.
- 2017 K. Koiliaris and C. Xu. A faster pseudopolynomial time algorithm for subset sum. In *Proceedings of the Twenty-Eighth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 1062–1072. SIAM, 2017.
- 2017 C. Chekuri and C. Xu. Computing minimum cuts in hypergraphs. In *Proceedings of the Twenty-Eighth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 1085–1100. SIAM, 2017.
- 2015 C. Chekuri, T. Rukkanchanunt, and C. Xu. On element-connectivity preserving graph simplification. In N. Bansal and I. Finocchi, editors, *Algorithms - ESA 2015*, volume 9294 of *Lecture Notes in Computer Science*, pages 313–324. Springer Berlin Heidelberg, 2015.
- 2015 H.-C. Chang, J. Erickson, and C. Xu. Detecting weakly simple polygons. In *Proceedings of the Twenty-Sixth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 1655–1670. SIAM, 2015.

Journal Publications

- 2025* J. Zhao, M. Xiao, and C. Xu. The Traveling Tournament Problem: Improved Algorithms Based on Cycle Packing. **Theoretical Computer Science**, page 115519, 2025.
- 2025 B. Cao and C. Xu. A Near-Linear Time Algorithm and a Min-Cost Flow Approach for Determining the Optimal Landing Times of a Fixed Sequence of Planes. **Annals of Operations Research**, 344(1):479–498, 2025.
- 2023 T. Hirayama, Y. Liu, K. Makino, K. Shi, and C. Xu. A polynomial time algorithm for finding a minimum 4-partition of a submodular function. **Mathematical Programming**, 2023.
- 2021 C. Beideman, K. Chandrasekaran, and C. Xu. Multicriteria cuts and size-constrained k -cuts in hypergraphs. **Mathematical Programming**, 2021.
- 2020 K.-i. Kawarabayashi and C. Xu. Minimum violation vertex maps and their applications to cut problems. **SIAM Journal on Discrete Mathematics**, 34(4):2183–2207, 2020.
- 2020 A. Gharehgozli, C. Xu, and W. Zhang. High multiplicity asymmetric traveling salesman problem with feedback vertex set and its application to storage/retrieval system. **European Journal of Operational Research**, 289(2):495–507, 2021.
- 2020 C. Chekuri, K. Quanrud, and C. Xu. LP Relaxation and Tree Packing for Minimum k -Cut. **SIAM Journal on Discrete Mathematics**, 34(2):1334–1353, 2020.
- 2021 A. Gharehgozli, C. Xu, and W. Zhang. High Multiplicity Asymmetric Traveling Salesman Problem with Feedback Vertex Set and Its Application to Storage/Retrieval System. **European Journal of Operational Research**, 289(2):495–507, 2021.
- 2019 K. Chandrasekaran, C. Xu, and X. Yu. Hypergraph k -cut in randomized polynomial time. **Mathematical Programming**, 186:85–113, March 2021.

- 2019 K. Koiliaris and C. Xu. Faster pseudopolynomial time algorithms for subset sum. **ACM Trans. Algorithms**, 15(3):40:1–40:20, June 2019.
- 2018 C. Chekuri and C. Xu. Minimum cuts and sparsification in hypergraphs. **SIAM Journal on Computing**, 47(6):2118–2156, 2018.
- 2018 C. Xu and Q. Zhang. The shortest kinship description problem. **Information Processing Letters**, 138:61 – 66, 2018.
- 2018 K. Bérczi, K. Chandrasekaran, T. Király, E. Lee, and C. Xu. Beating the 2-approximation factor for global bicut. **Mathematical Programming**, 177(1):291–320, Sep 2019.
- 2016 C. Xu. Reconstructing edge-disjoint paths faster. **Operations Research Letters**, 44(2):174 – 176, 2016.
- 2013 N. J. Calkin, J. E. Janoski, A. Nelson, S. Ryan, and C. Xu. Champion spiders in the game of Graph Nim. **Congr. Numer.**, 218:5–19, 2013.

Teaching

- S 2022 CS @ UESTC.
- F 2016 CS 374 Algorithms and Models of Computation @ UIUC. Teaching Assistant
- F 2015 CS 498 DL1 “new ” CS 473 Theory II @ UIUC. Teaching Assistant
- S 2015 CS 498 DL1 “new ” CS 473 Theory II @ UIUC. Teaching Assistant
- F 2014 CS 374 Algorithms and Models of Computation @ UIUC. Teaching Assistant
- F 2013 CS 373 Introduction to Theory of Computation @ UIUC. Teaching Assistant
- F 2010 AMS 345 Computational Geometry @ Stony Brook University. Teaching Assistant